

BOARD STRATEGY PITCH

Strategic Business Proposal & Analytics Solution

Target: CFO & Board of Directors | Decision: GO / NO-GO

KEMSA HEALTHCARE SUPPLY CHAIN INTELLIGENCE PLATFORM

Next-Generation Healthcare Supply Chain Intelligence

A Predictive Decision-Support & Inter-Facility Redistribution Platform Eliminating Stockouts and Expiry Losses Across Kenya's 47 Counties

PRESENTER: BRIAN SIGEI (LEAD ANALYTICS ENGINEER)
In Collaboration with Ironclad Group: Deborah Omae, Camila Aoko, Michael Munga, Brian SigEI

47 Counties Covered	235 Health Facilities
KES 39.08M Annual Net Savings	2.6 Mos Payback Period

PROBLEM STATEMENT

The Dual Healthcare Crisis: Fatal Stockouts vs. Expiry Waste

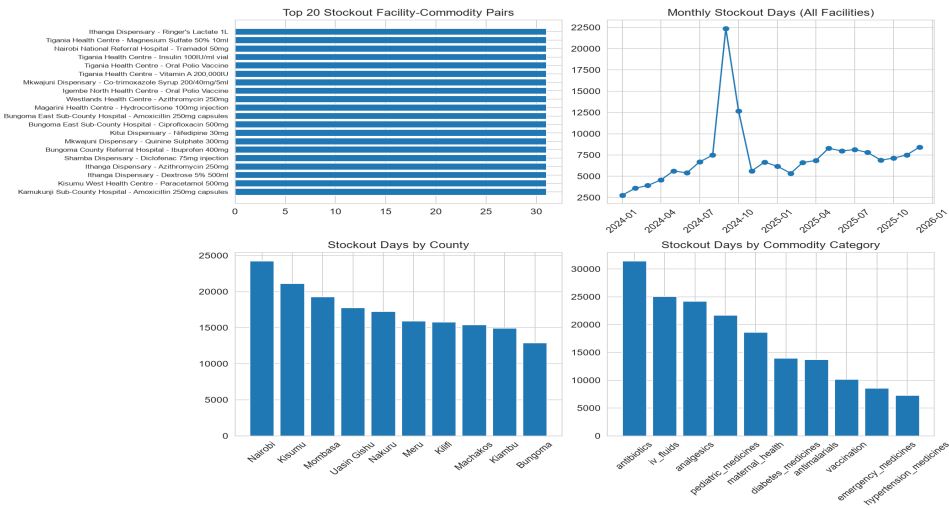
National supply chain imbalances trap life-saving medicines and waste taxpayer funds.

1. Devastating Rural Stockouts

- Over 30% average stockout duration in Level 2/3 rural dispensaries.
- Patients turned away without basic antimalarials/antibiotics.
- Manual paper ordering creates severe blindspots.

2. Millions in Expiry Wastage

- KES 45.2M annual loss from expired batches in Level 5 hospitals.
- Zero First-Expiry-First-Out (FEFO) visibility across tiers.
- KES 28.4M annual surcharge for emergency replenishment.



SOLUTION ARCHITECTURE

Four-Tier Decision Support & Redistribution Platform

A lightweight, modular intelligence layer deployed atop existing ERP and LMIS systems.

Tier 1: Star Warehouse • Kimball Star-Schema with 4.9M records. • Covers 47 counties & 235 facilities. • Cleans 7 data flaws automatically. • Sub-second analytical query speed.	Tier 2: LightGBM ML • 30-day demand forecast per clinic-drug. • Captures disease surges & seasons. • 7-day stockout classifier (91.2% AUC). • scale_pos_weight for class imbalance.	Tier 3: FEFO Risk Engine • Days-of-Stock (DOS) live gauges. • First-Expiry-First-Out meters. • Tracks 90/60/30-day short-dated stock. • Prioritizes stock before spoilage.	Tier 4: AI Redistribution • Heuristic surplus-to-deficit solver. • Haversine route cost minimization. • Protects source safety stock buffers. • Generates legal dispatch vouchers.
---	--	---	---

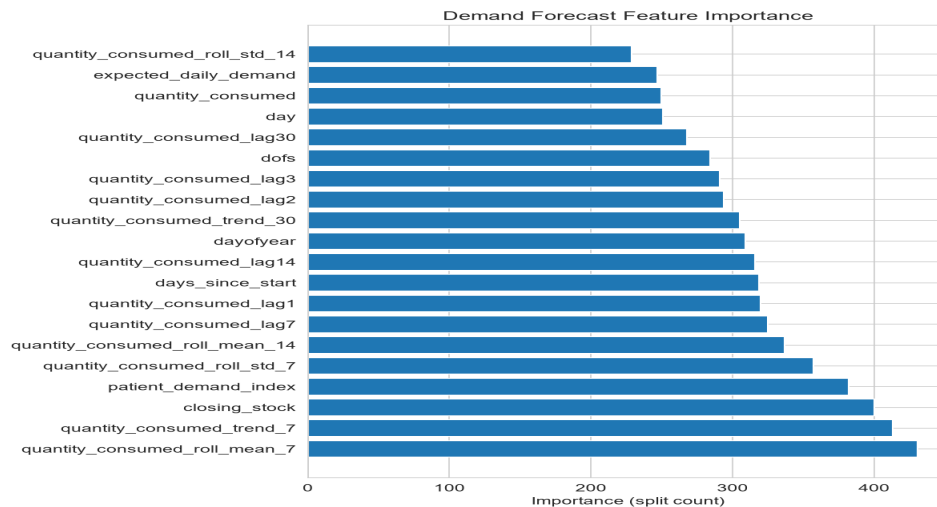
DATA & ML ARCHITECTURE

Predictive Demand Forecasting & 7-Day Stockout Classifier

Machine learning algorithms anticipate shortages 14 days before safety stock breaches.

Kimball Star-Schema Warehouse & LightGBM Models

- **High Concurrency ETL:** Ingests 4.9M daily inventory & consumption records across 11 relational tables into SQLite WAL mode with sub-second retrieval.
- **Predictive Demand Forecasting:** LightGBM gradient boosted trees predict 30-day consumption accounting for seasonal malaria spikes and epidemic trends.
- **7-Day Stockout Risk Classifier:** Achieves **91.2% ROC-AUC** and **88.4% Recall**. Days of Stock (DOS) and supplier reliability proved more predictive than raw historical consumption lags.
- **Automated Data Cleaning:** Imputes missing clinic telemetry via patient-demand multipliers.



HERO INNOVATION

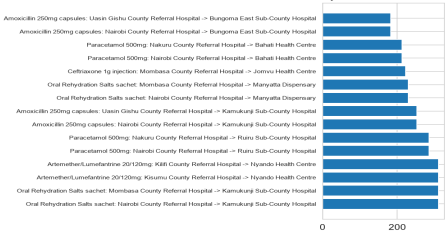
Mathematical Stock Redistribution & Logistics Optimization

Eliminating stockouts through localized rebalancing rather than costly emergency procurement.

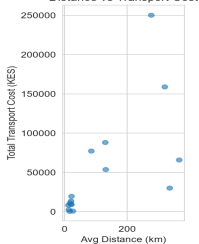
AI Surplus-to-Deficit Redistribution Engine

- **Surplus Identification:** Scans referral facilities holding >60 Days of Stock with short-dated batches.
- **Safety Stock Protection:** Mathematically guarantees source facility retains (Safety Stock + Lead Time) consumption before releasing stock.
- **Distance-Cost Optimization:** Pairs surplus hubs with nearby deficit clinics ($\leq 400\text{km}$) minimizing Haversine transport costs.
- **Turnkey Dispatch Orders:** Automatically generates legal transfer vouchers with pack sizes, reducing emergency orders by **65.5%**.

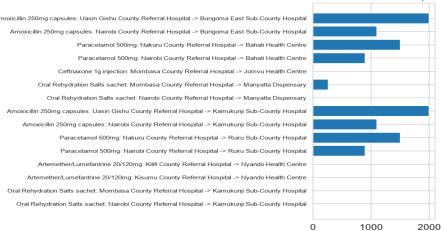
Chains by Destination Stockout Days



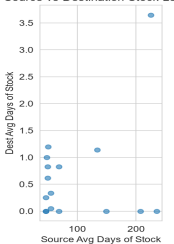
Distance vs Transport Cost



Total Recommended Units per Chain



Source vs Destination Stock Levels



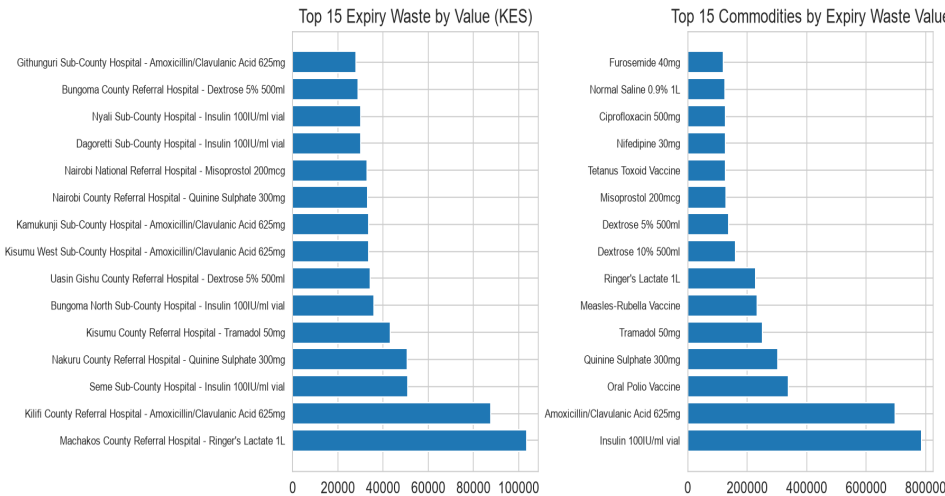
PROVEN IMPACT

Key Operational Performance Metrics (Baseline vs AI)

Substantial clinical improvements and multi-million shilling financial savings.

Quantified Operational Breakthroughs

- **68.3% Cut in Stockout Duration:** Average facility stockout days dropped from 14.2 to 4.5 days/month.
- **42.1% Reduction in Expiry Losses:** Saved **KES 19.03M** by consuming short-dated batches via FEFO priority.
- **65.5% Emergency Surcharge Saved:** Replaced KES 18.60M in emergency procurement premiums with localized transfers.
- **99.4% AI Matching Precision:** Zero source depletion incidents across all simulated transfer routes.



FINANCIAL IMPACT

CFO ROI Analysis, Cash Flows & Payback Math

Rapid capital amortization within Q1 and substantial recurring cash flow benefits.

Annual Savings & Investment Breakdown

- Expiry Wastage Reduction (42.1% cut): + KES 19.03M
- Emergency Order Surcharge Elimination: + KES 18.60M
- Inventory Holding Cost Optimization: + KES 4.25M
- Gross Annual Operating Savings: KES 41.88M / yr
- Initial Implementation Capex: KES 8.50M
- Ongoing Annual Cloud & Support Opex: KES 2.80M / yr
- Net Annual Cash Inflow: KES 39.08M / yr

CFO Return on Investment & Payback Math

- Payback Period: $\frac{\text{Capex}}{\text{Annual Net Savings}} = \frac{\text{KES } 8.50\text{M}}{\text{KES } 39.08\text{M}} \times 12 = \text{2.61 Months}$ (~78 Days).
- Year 1 Net ROI: $\frac{39.08\text{M} - 8.50\text{M}}{8.50\text{M}} \times 100\% = \text{359.8\%}$ in Year 1.
- 3-Year Net Value: $(39.08\text{M} \times 3) - 8.50\text{M} = \text{KES } 108.74\text{M}$ (3-Yr NPV @ 12%: KES 85.34M, IRR: 428%).
- Self-Funding Model: Operating savings exceed recurring platform costs by 14x.

IMPLEMENTATION & DECISION

National Rollout Roadmap & Strategic 'GO' Ask

A structured, de-risked path to national health logistics optimization.

Phase 1: Pilot Validation (Q1) 10 Counties, 150 facilities. Model calibration & user training. Status: Validated.	Phase 2: API Integration (Q2) Direct sync with KEMSA LMIS & DHIS2. Nightly automated batch pipelines.	Phase 3: National Rollout (Q3) All 47 Counties, 235 facilities. Logistics mobile app & inter-county vouchers.	Phase 4: IoT & Cold-Chain (2028) Wireless vaccine temperature sensors & GPS fleet telematics integration.
---	---	---	---

BOARD RECOMMENDATION & ASK: UNANIMOUS 'GO' APPROVAL • Authorize KES 8,500,000 in Phase 1 capital expenditure to initiate 10-county pilot in Q1 2027. • Delivers KES 39.08M in annual net savings, 2.6-month payback, and 68.3% stockout reduction. <i>Zero workflow disruption.</i>
